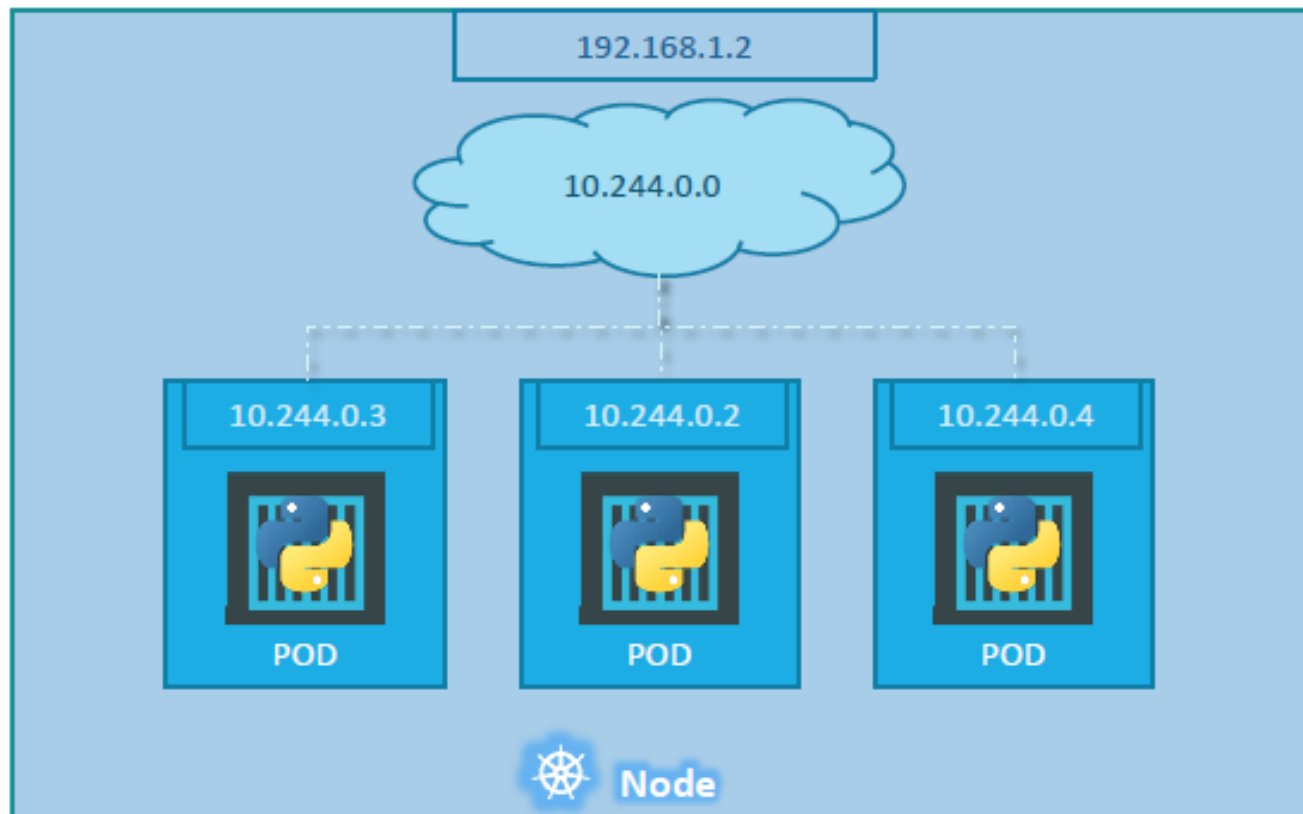


Kubernetes networking



Pod network

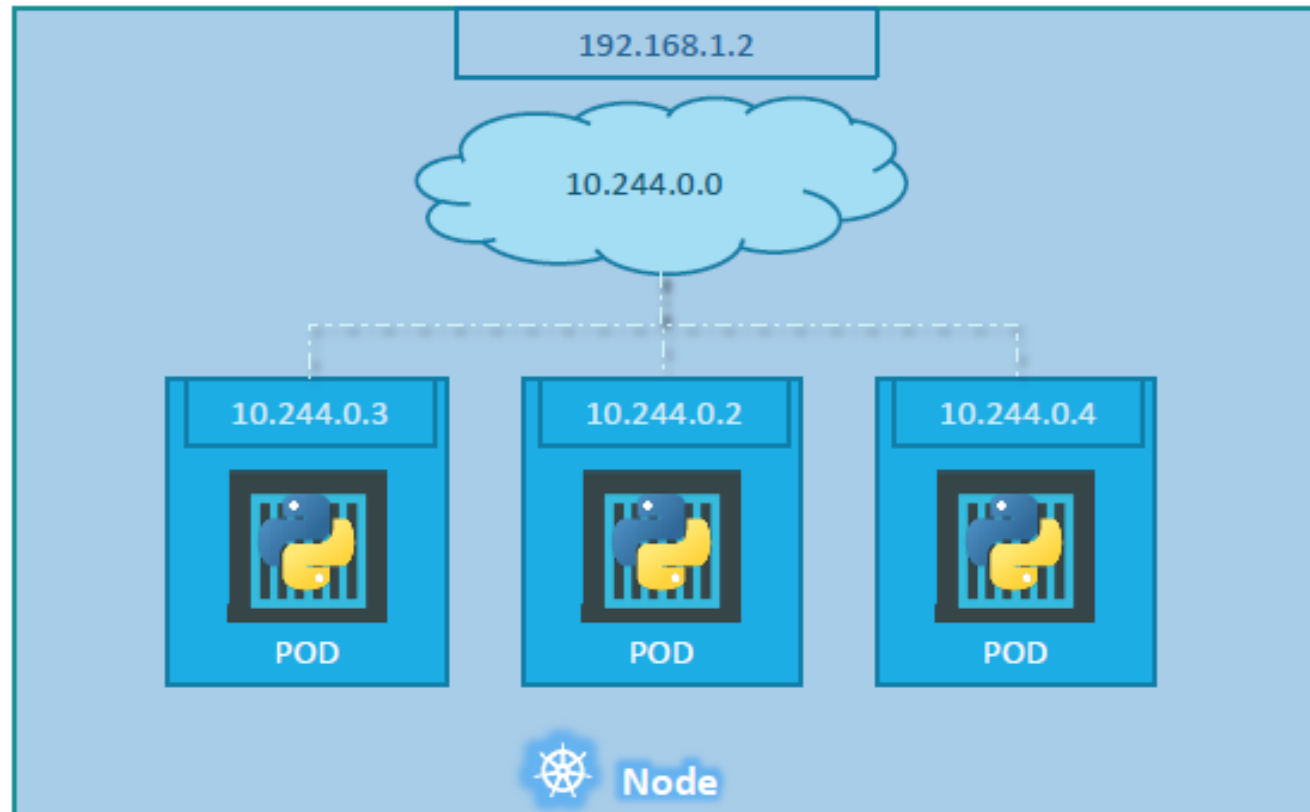


Kubernetes networks

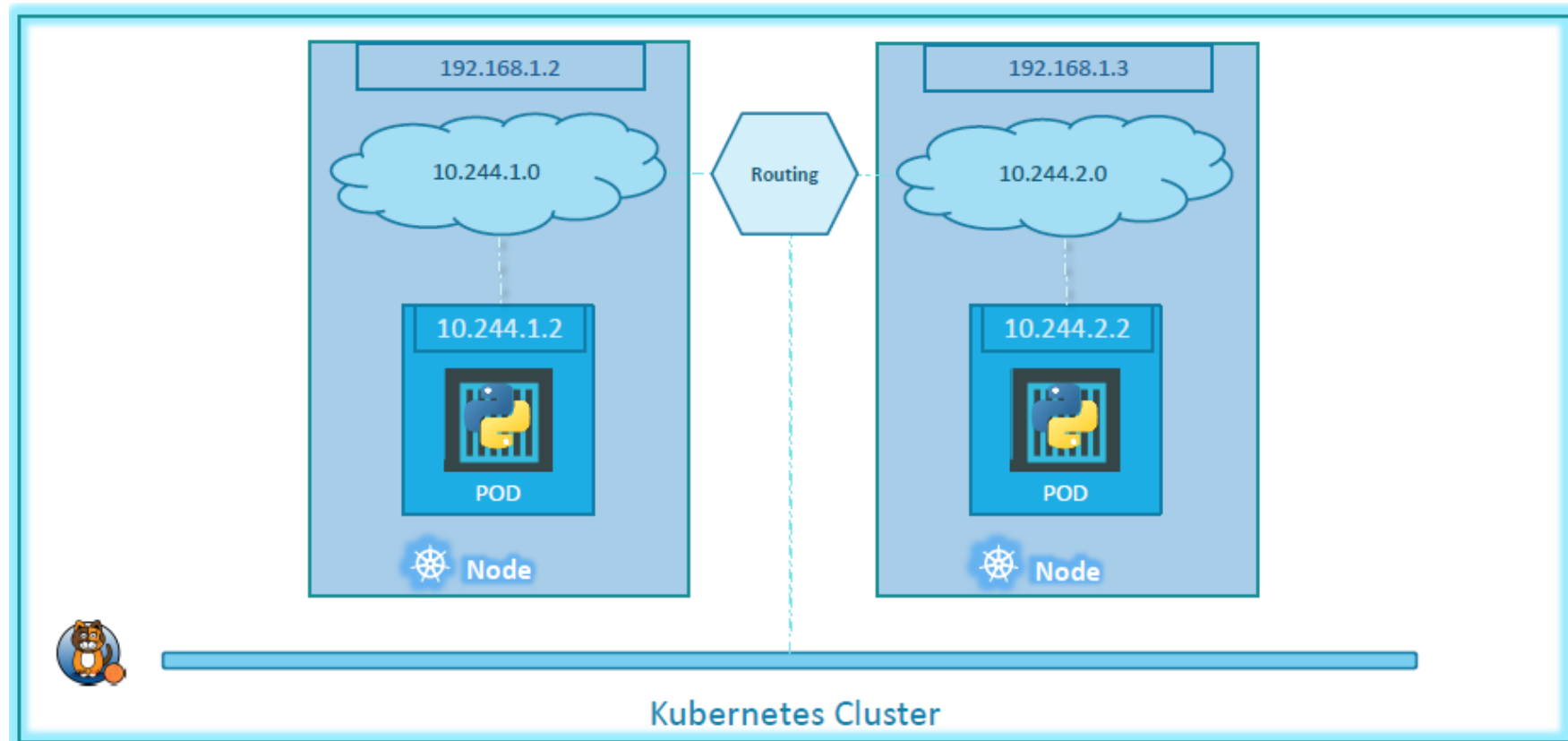
Nodes network: la rete a cui sono collegati i nodi del cluster (es: 192.168.1.0/24)

Pod network: la rete (software-defined) su cui vengono attaccati i Pod (es: 10.244.0.0/16)

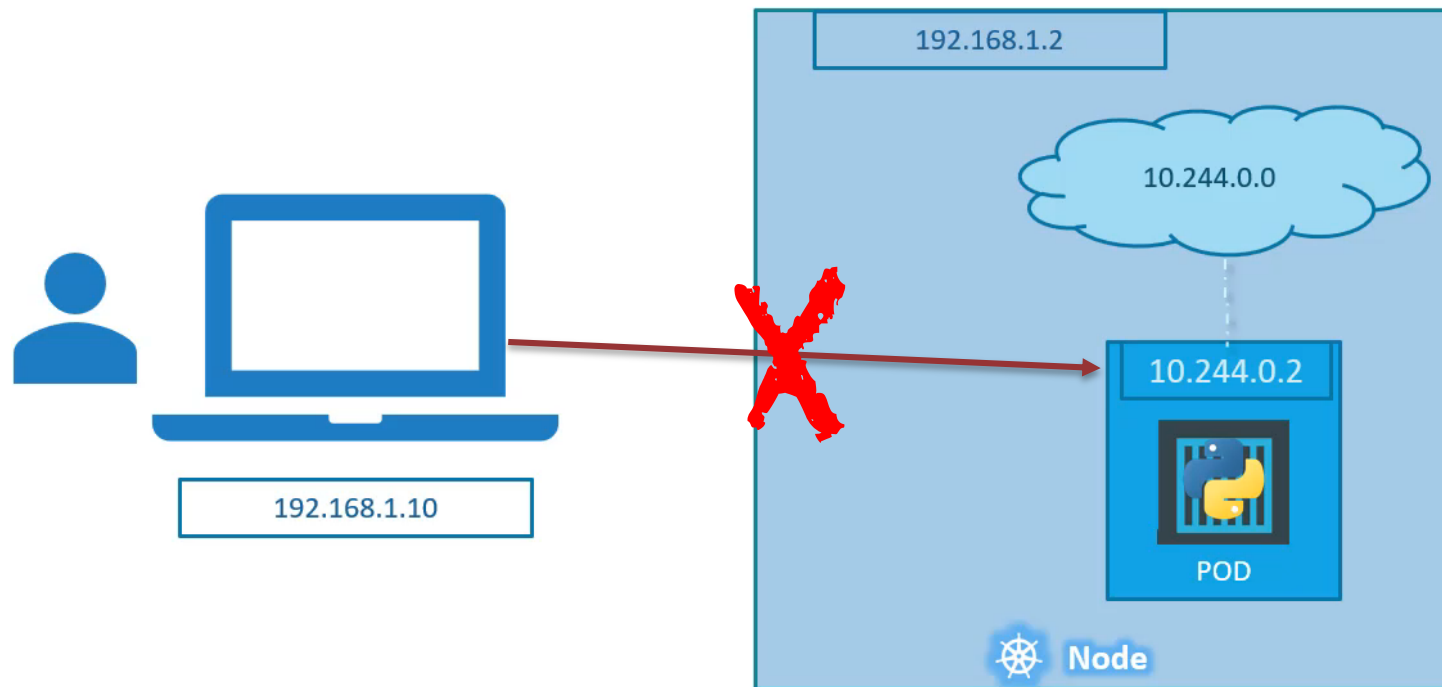
Pod network single node



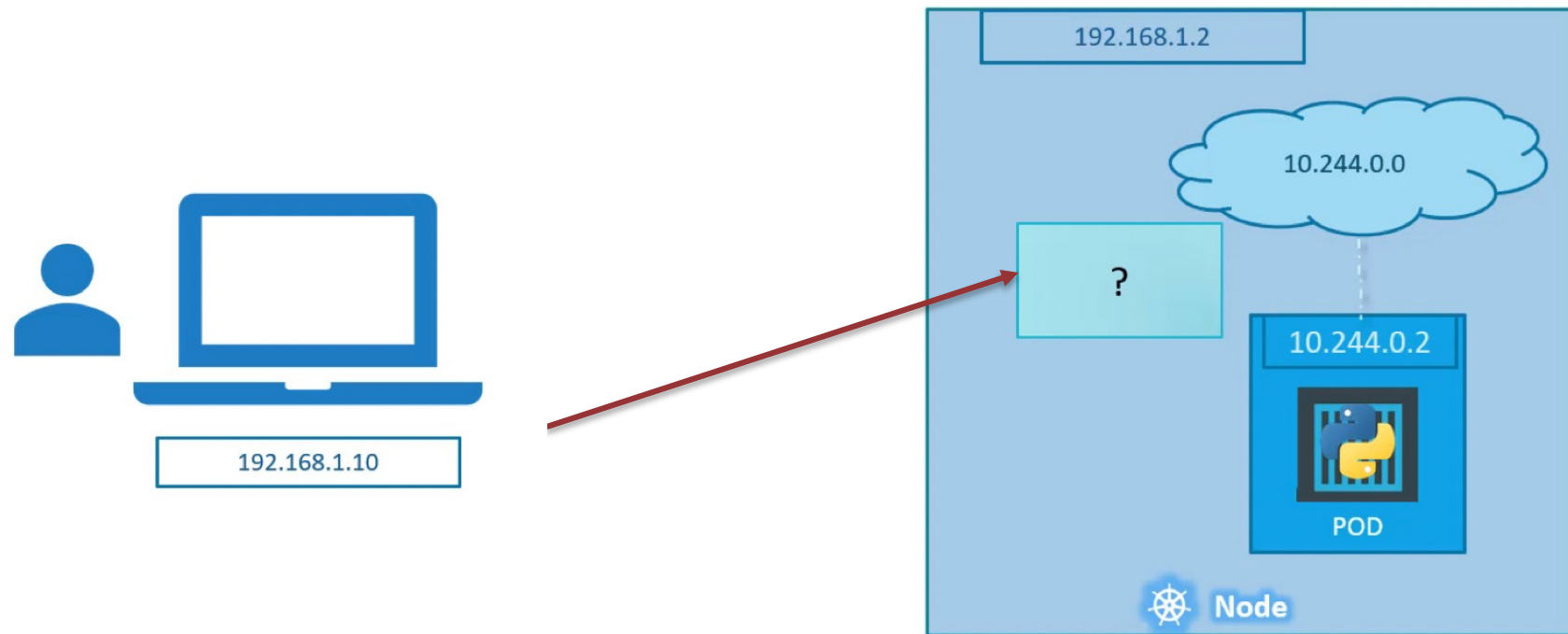
Pod network multi node



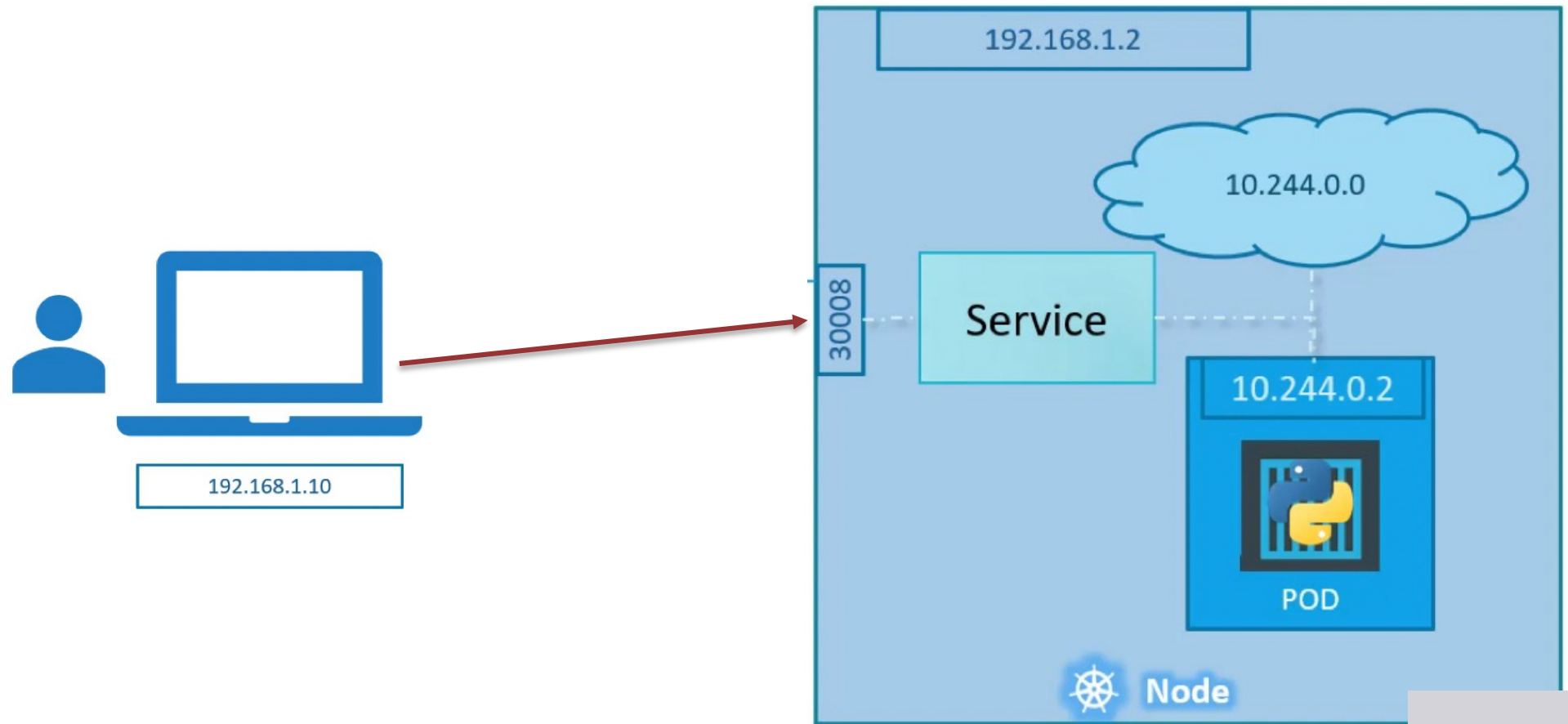
Pod network



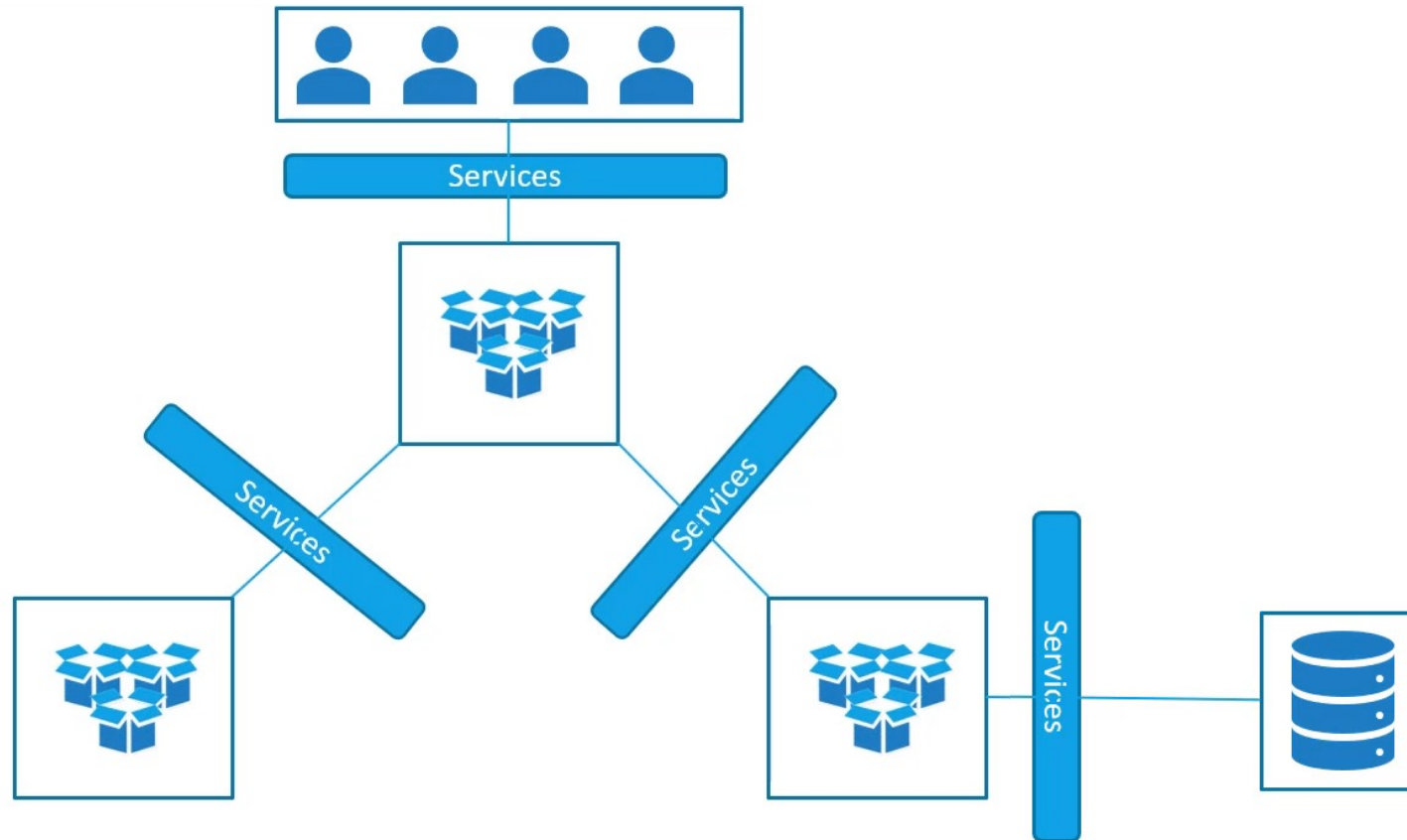
Come raggiungere un Pod



Come raggiungere un Pod



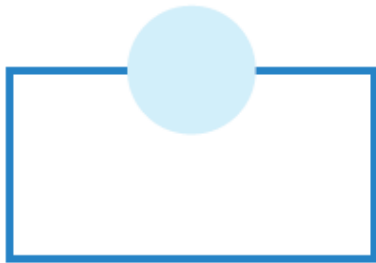
Services



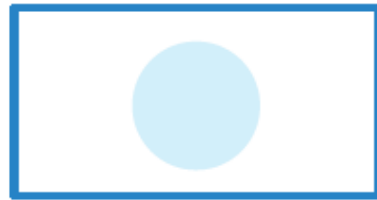
Services network

- Service network: la rete (software-defined) su cui vengono attaccati i Service (es: 10.96.0.0/16)

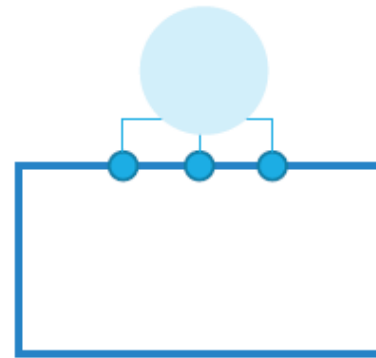
Tipologie di Services



NodePort

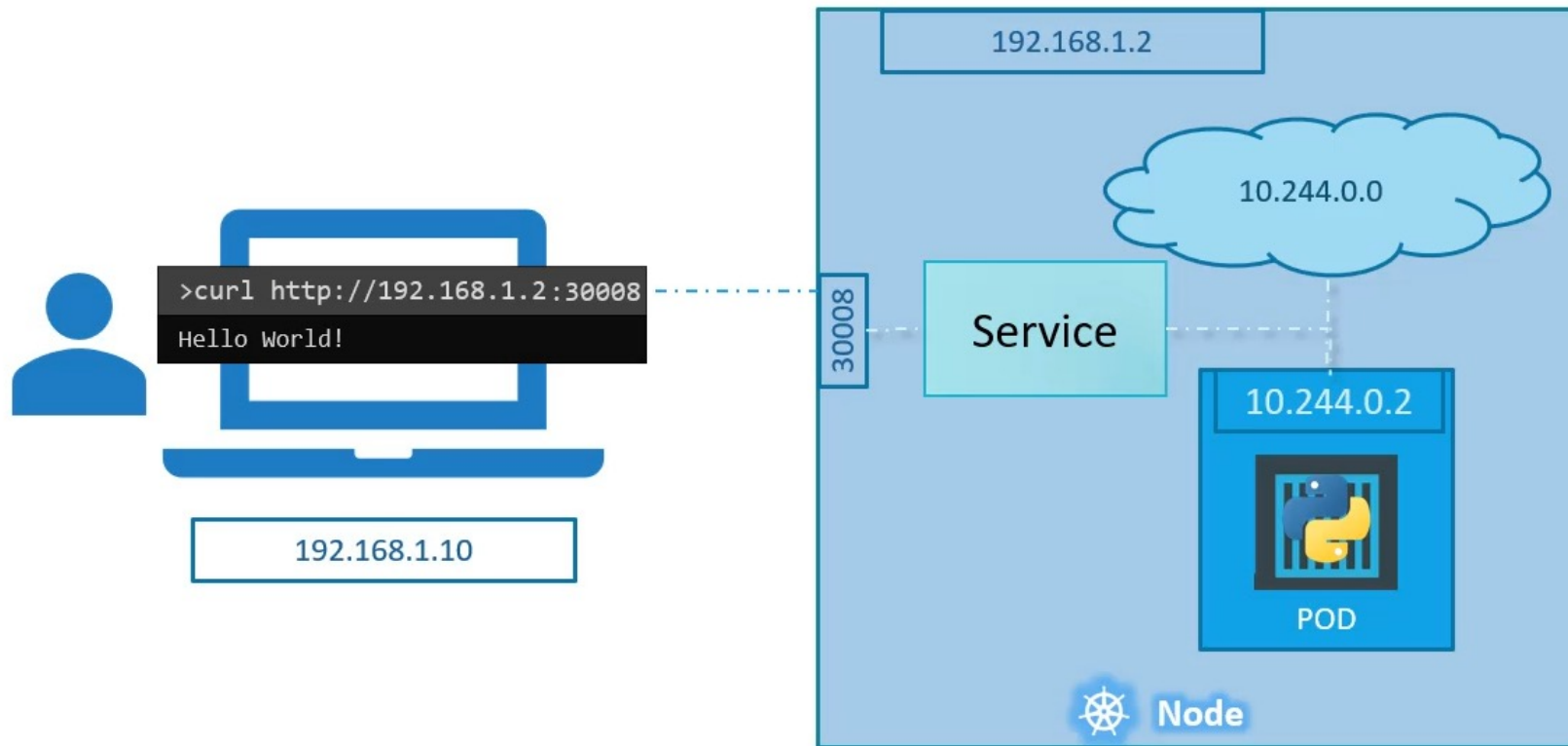


ClusterIP



LoadBalancer

Service NodePort

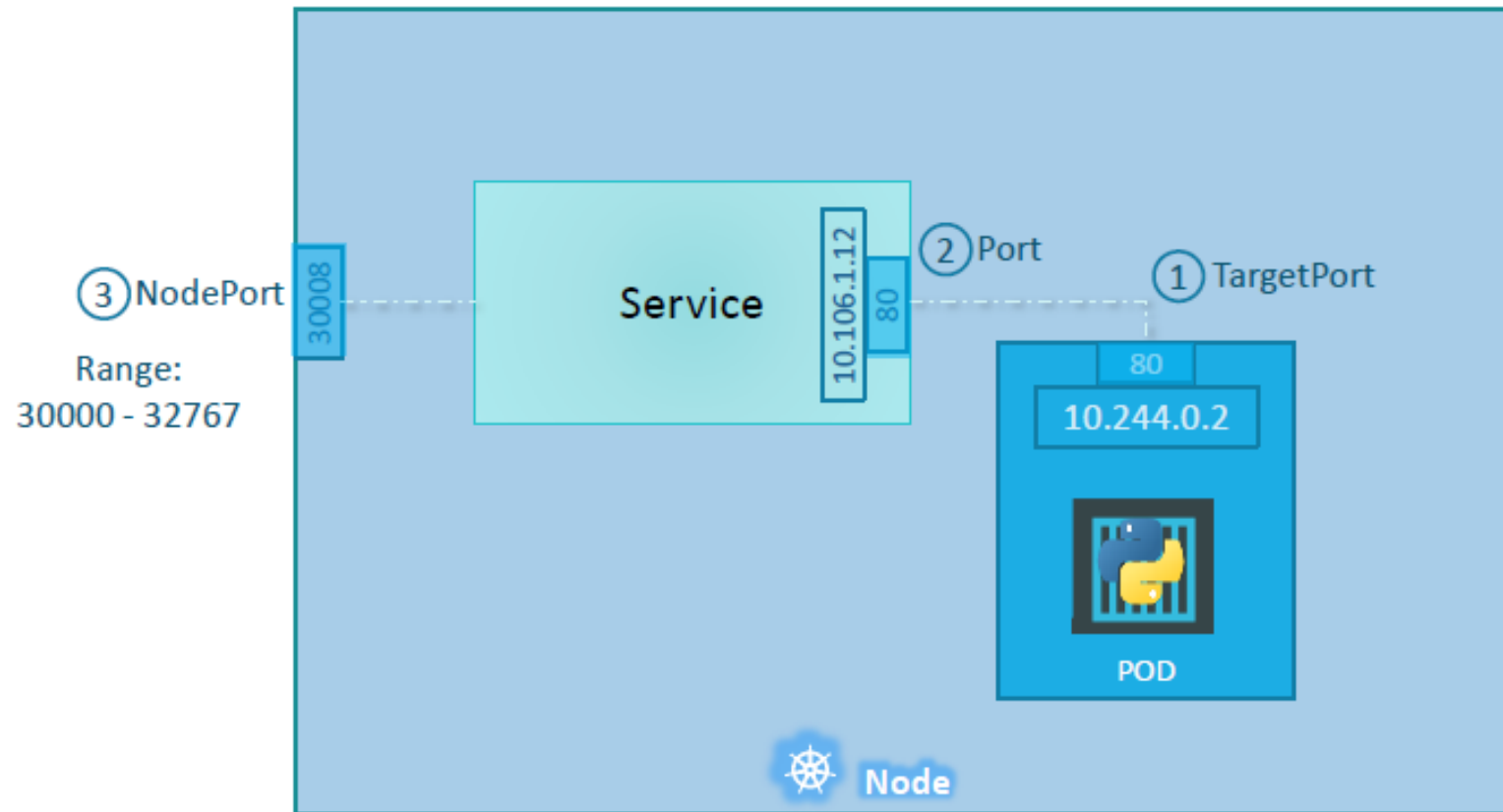


Service NodePort

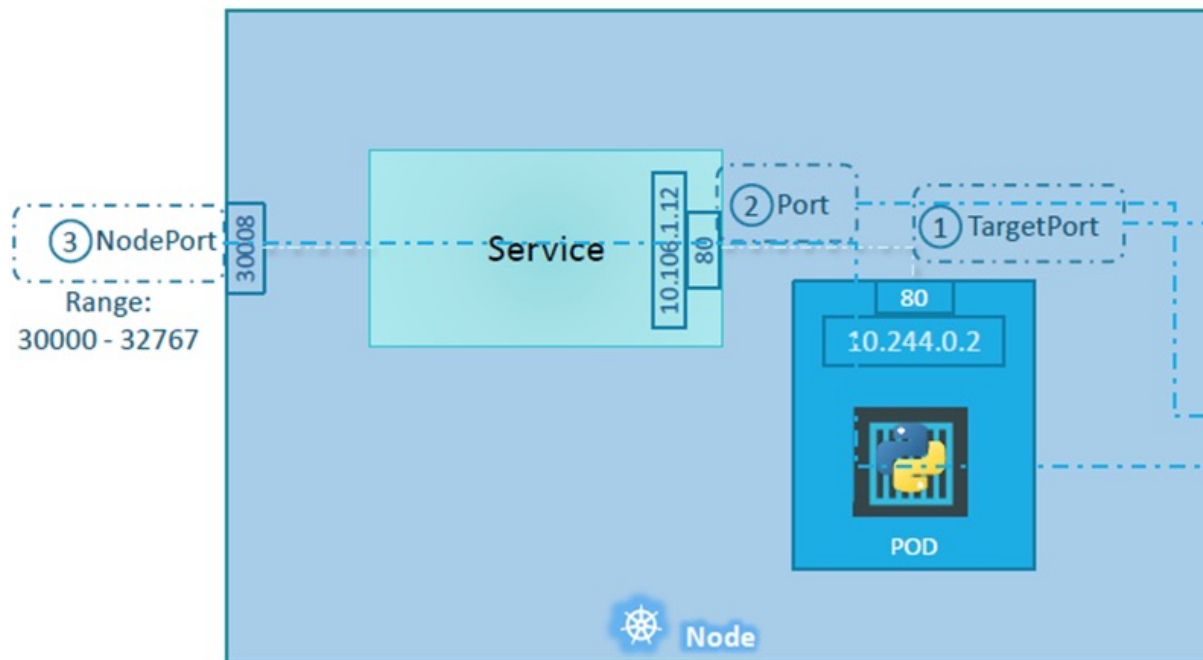
Servono per esporre i Pod al do fuori del cluster Kubernetes.

Aprono una porta su tutti i nodi e la reindirigono sulla porta aperta dai Pod.

Service NodePort



Service NodePort



```
service-definition.yml
apiVersion: v1
kind: Service
metadata:
  name: myapp-service
spec:
  type: NodePort
  ports:
    - targetPort: 80
      port: 80
      nodePort: 30008
```

Service NodePort

service-definition.yml

```
apiVersion: v1
kind: Service
metadata:
  name: myapp-service
spec:
  type: NodePort
  ports:
    - targetPort: 80
      port: 80
      nodePort: 30008
  selector:
    app: myapp
    type: front-end
```

pod-definition.yml

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  labels:
    app: myapp
    type: front-end
spec:
  containers:
    - name: nginx-container
      image: nginx
```

Service NodePort

```
> kubectl create -f service-definition.yml  
service "myapp-service" created
```

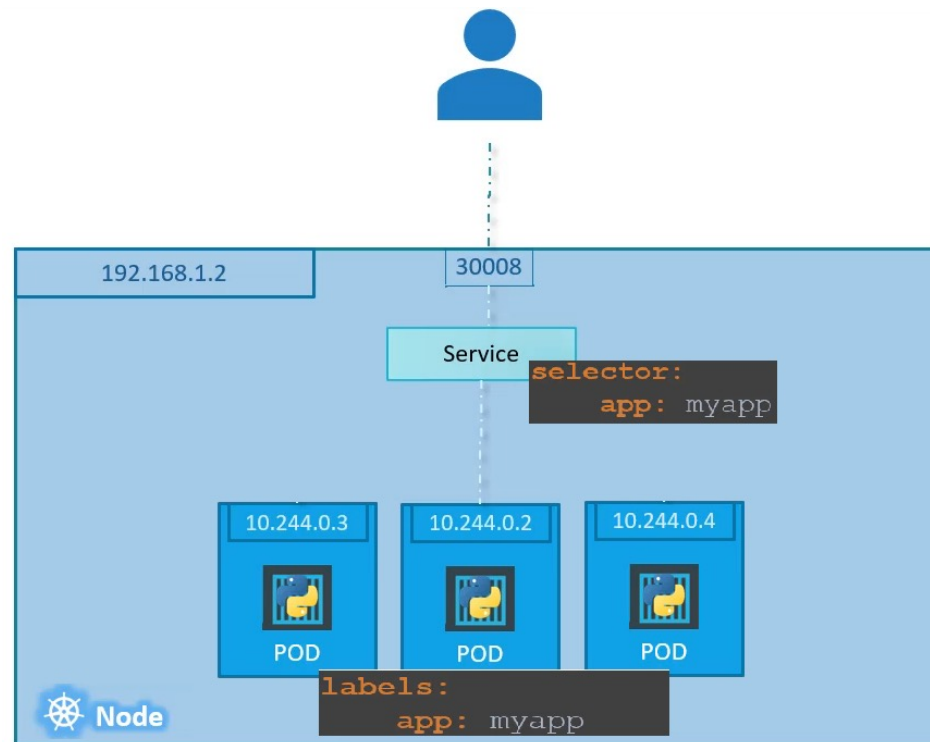
```
> kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	16d
myapp-service	NodePort	10.106.127.123	<none>	80:30008/TCP	5m

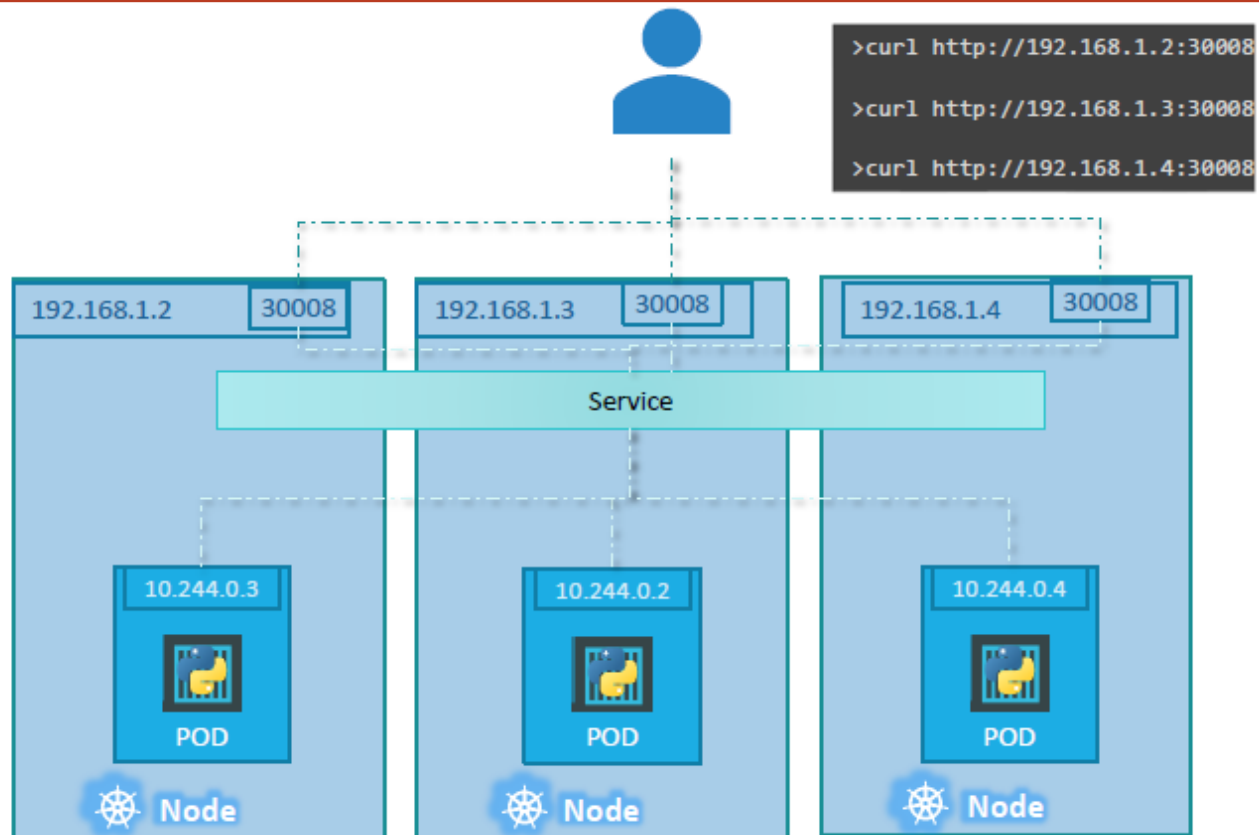
```
> curl http://192.168.1.2:30008
```

```
<html>  
<head>  
<title>Welcome to nginx!</title>  
<style>  
  body {  
    width: 35em;  
    margin: 0 auto;  
    font-family: Tahoma, Verdana, Arial, sans-serif;  
  }  
</style>  
</head>  
<body>
```

Service NodePort

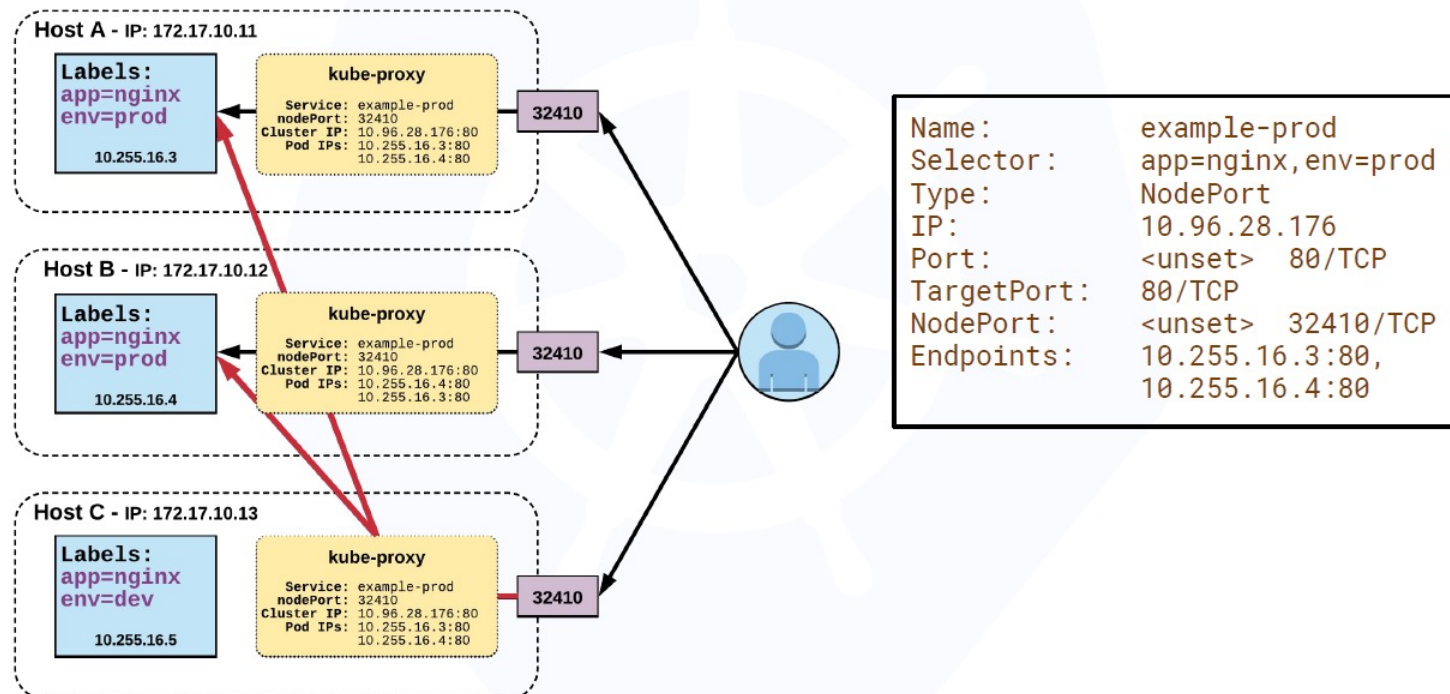


Service NodePort



Service NodePort

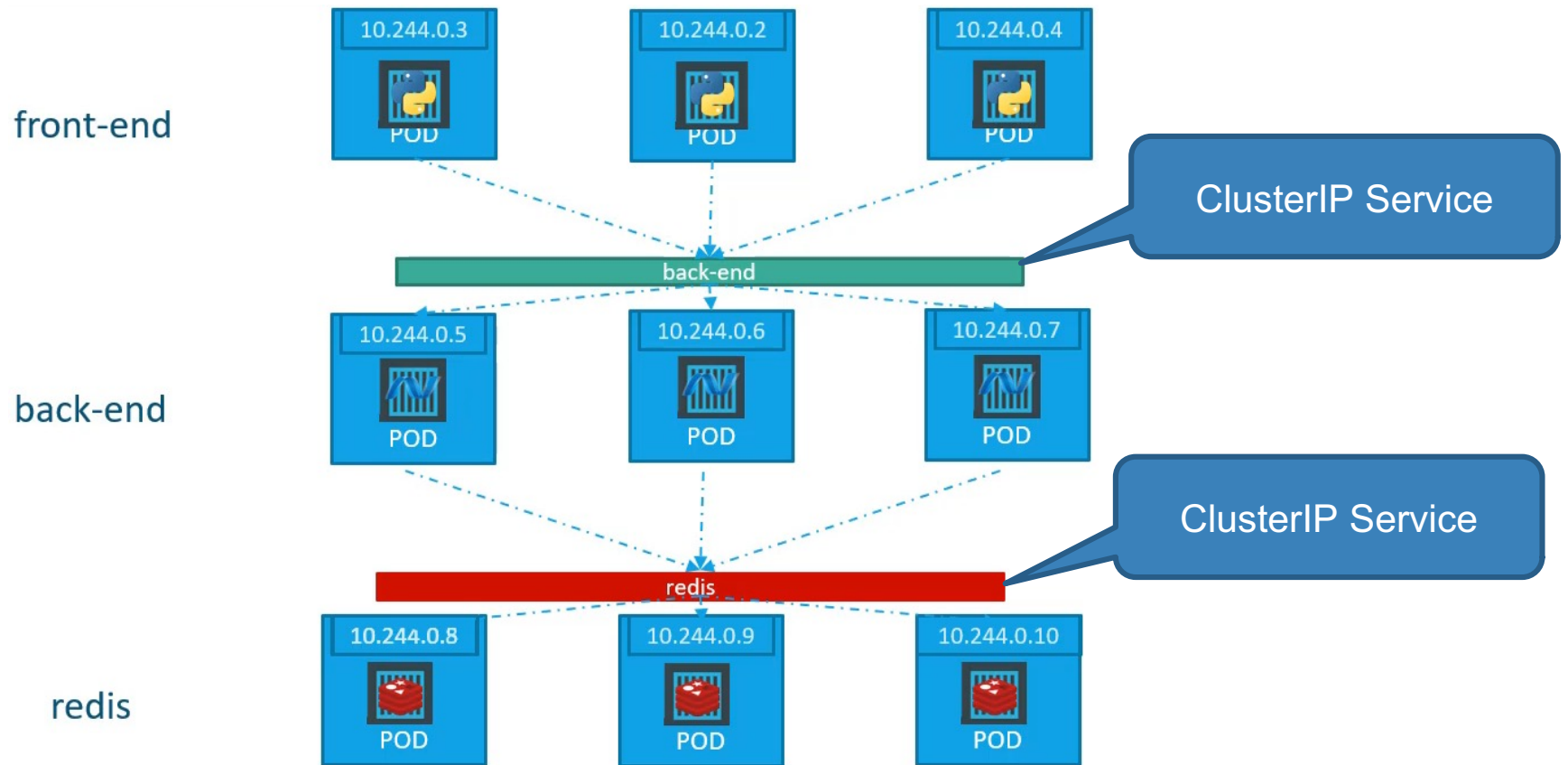
NodePort Service



Service ClusterIP

Servono per fare comunicare i Pod fra di loro, non sono raggiungibili dall'esterno del cluster.

Service ClusterIP



Service ClusterIP

service-definition.yml

```
apiVersion: v1
kind: Service
metadata:
  name: back-end
spec:
  type: ClusterIP
  ports:
    - targetPort: 80
      port: 80
  selector:
    app: myapp
    type: back-end
```

pod-definition.yml

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  labels:
    app: myapp
    type: back-end
spec:
  containers:
    - name: nginx-container
      image: nginx
```



Service ClusterIP

```
> kubectl create -f service-definition.yml
```

```
service "back-end" created
```

```
> kubectl get services
```

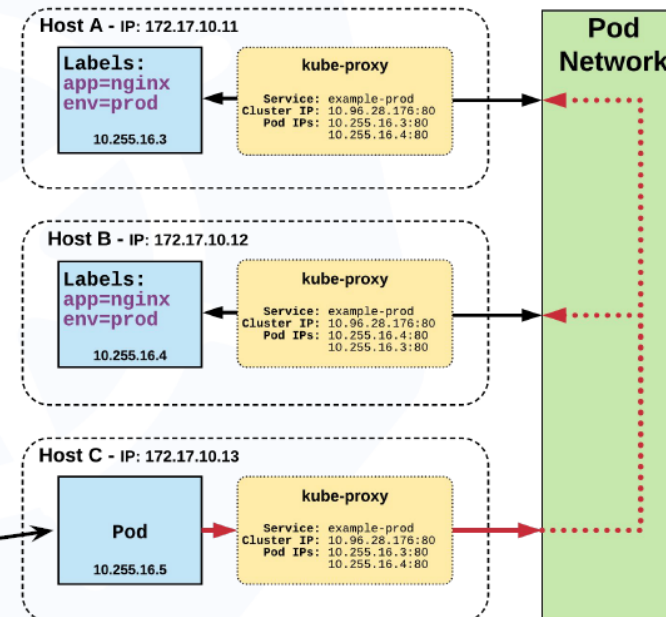
NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
back-end	ClusterIP	10.106.127.123	<none>	80/TCP	2m

Service ClusterIP

Cluster IP Service

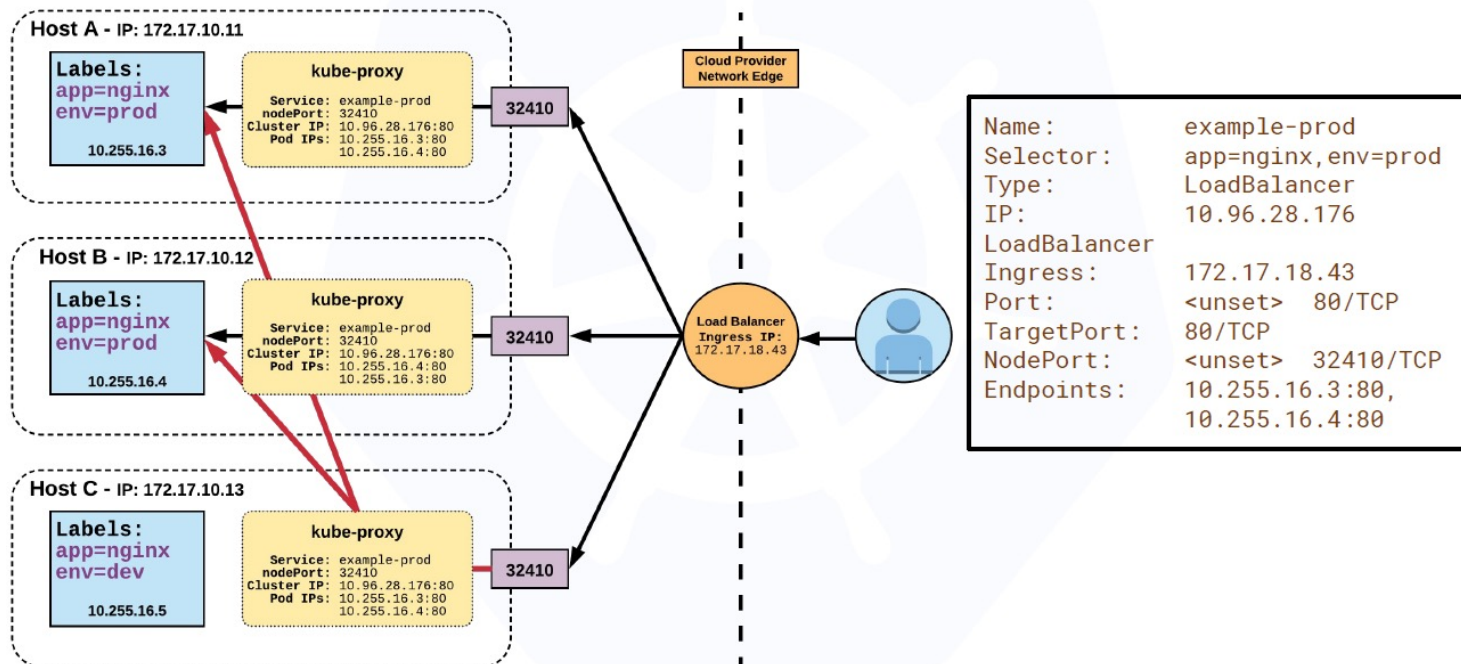
```
Name:      example-prod
Selector:   app=nginx,env=prod
Type:       ClusterIP
IP:         10.96.28.176
Port:       <unset> 80/TCP
TargetPort: 80/TCP
Endpoints:  10.255.16.3:80,
            10.255.16.4:80
```

```
/ # nslookup example-prod.default.svc.cluster.local
Name:      example-prod.default.svc.cluster.local
Address 1: 10.96.28.176 example-prod.default.svc.cluster.local
```

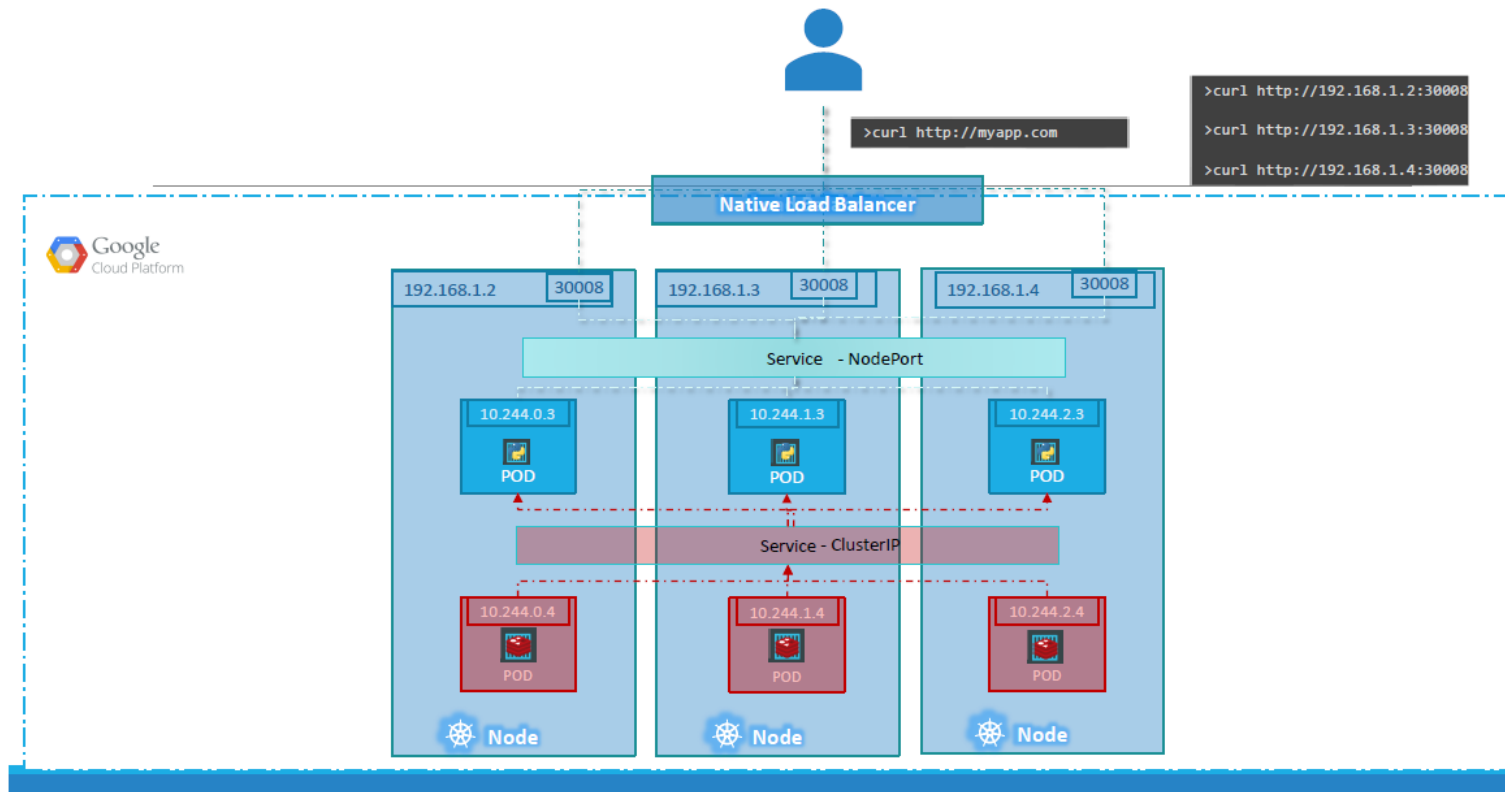


Service LoadBalancer

LoadBalancer Service



Services



Services

service-definition.yml

```
apiVersion: v1
kind: Service
metadata:
  name: front-end
spec:
  type: NodeBalancer
  ports:
    - targetPort: 80
      port: 80
  selector:
    app: myapp
    type: front-end
```

```
> kubectl create -f service-definition.yml
```

```
service "front-end" created
```

```
> kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	16d
front-end	LoadBalancer	10.106.127.123	<Pending>	80/TCP	2m

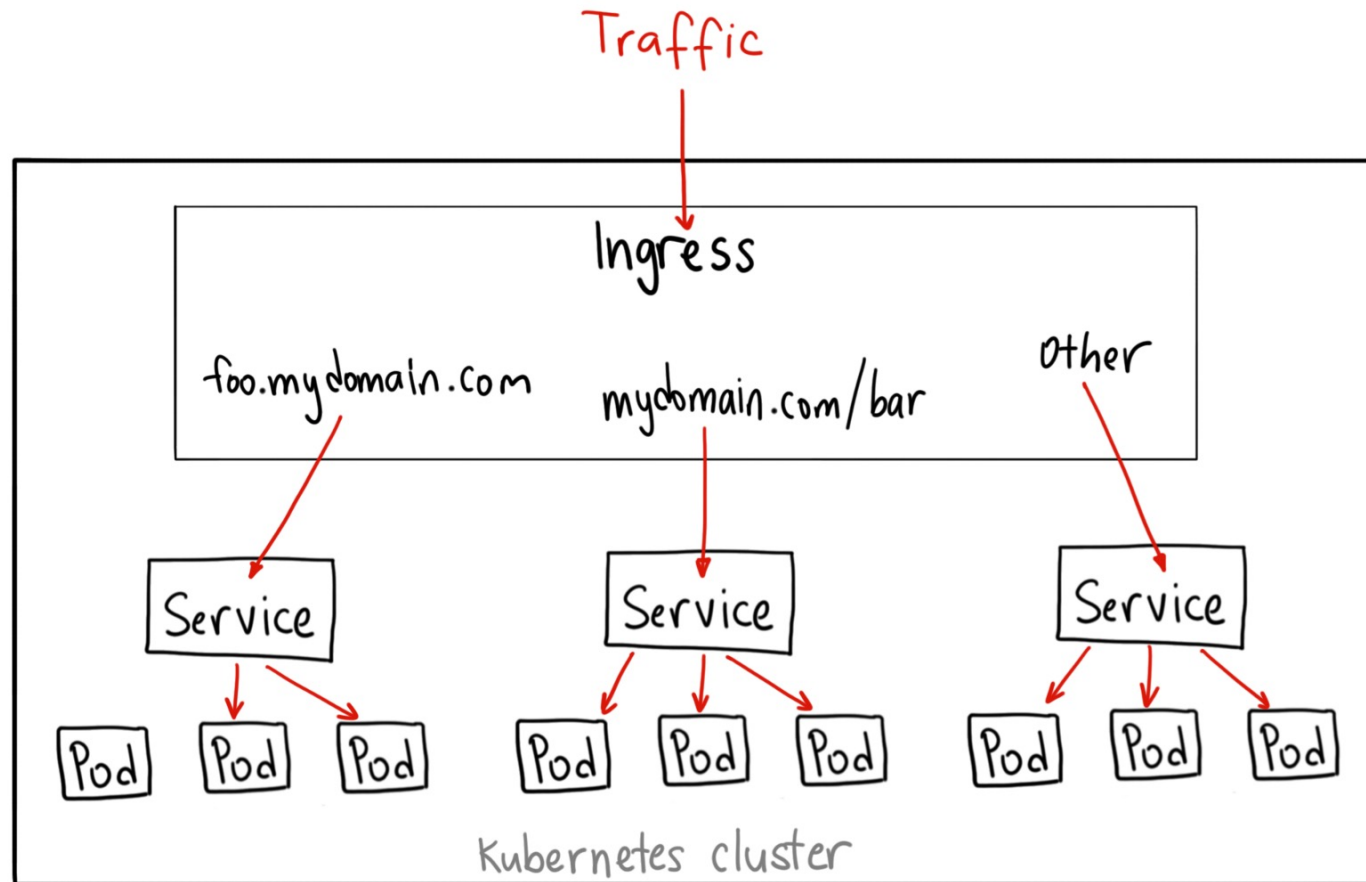
LAB

https://github.com/sunnyvale-academy/ITS-ICT_Containers

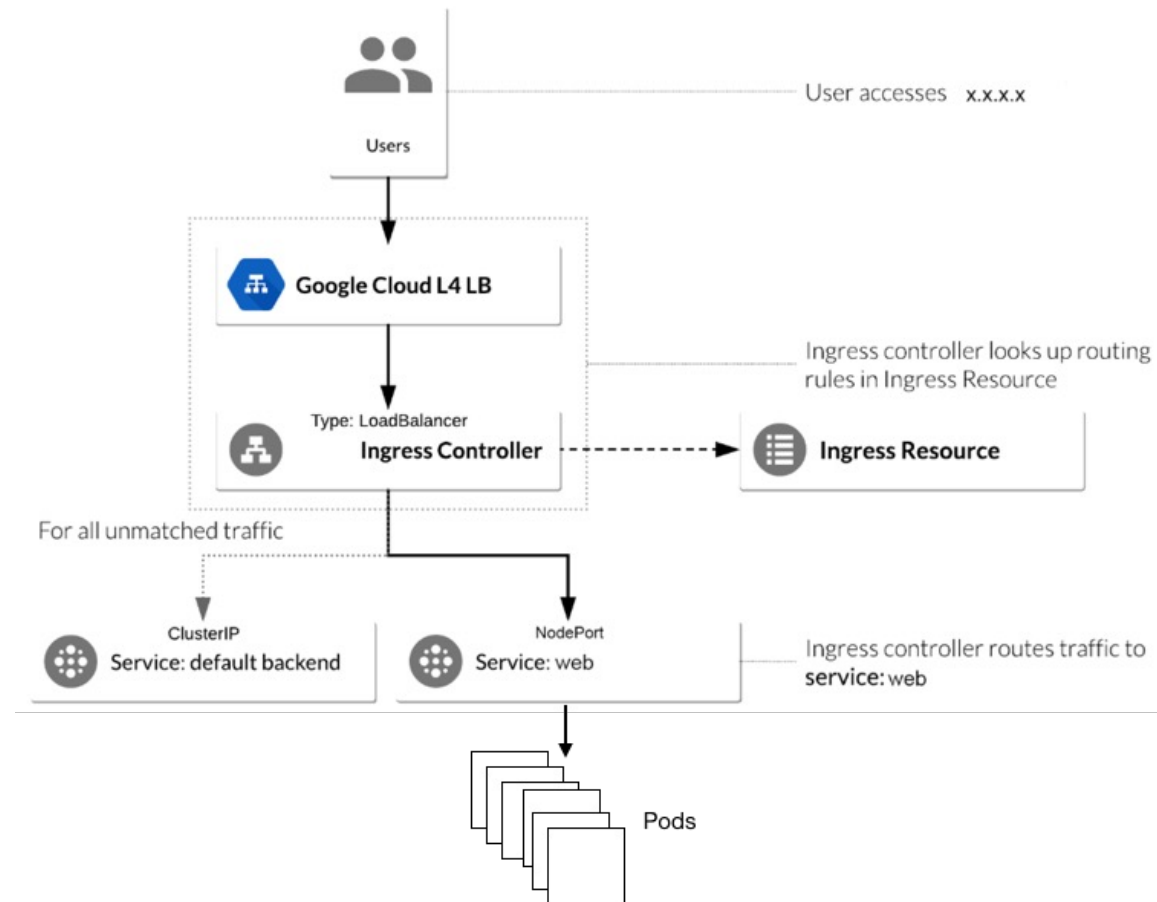
Lab 23 – Service

Assignment 06 - Expose Deployments using Services

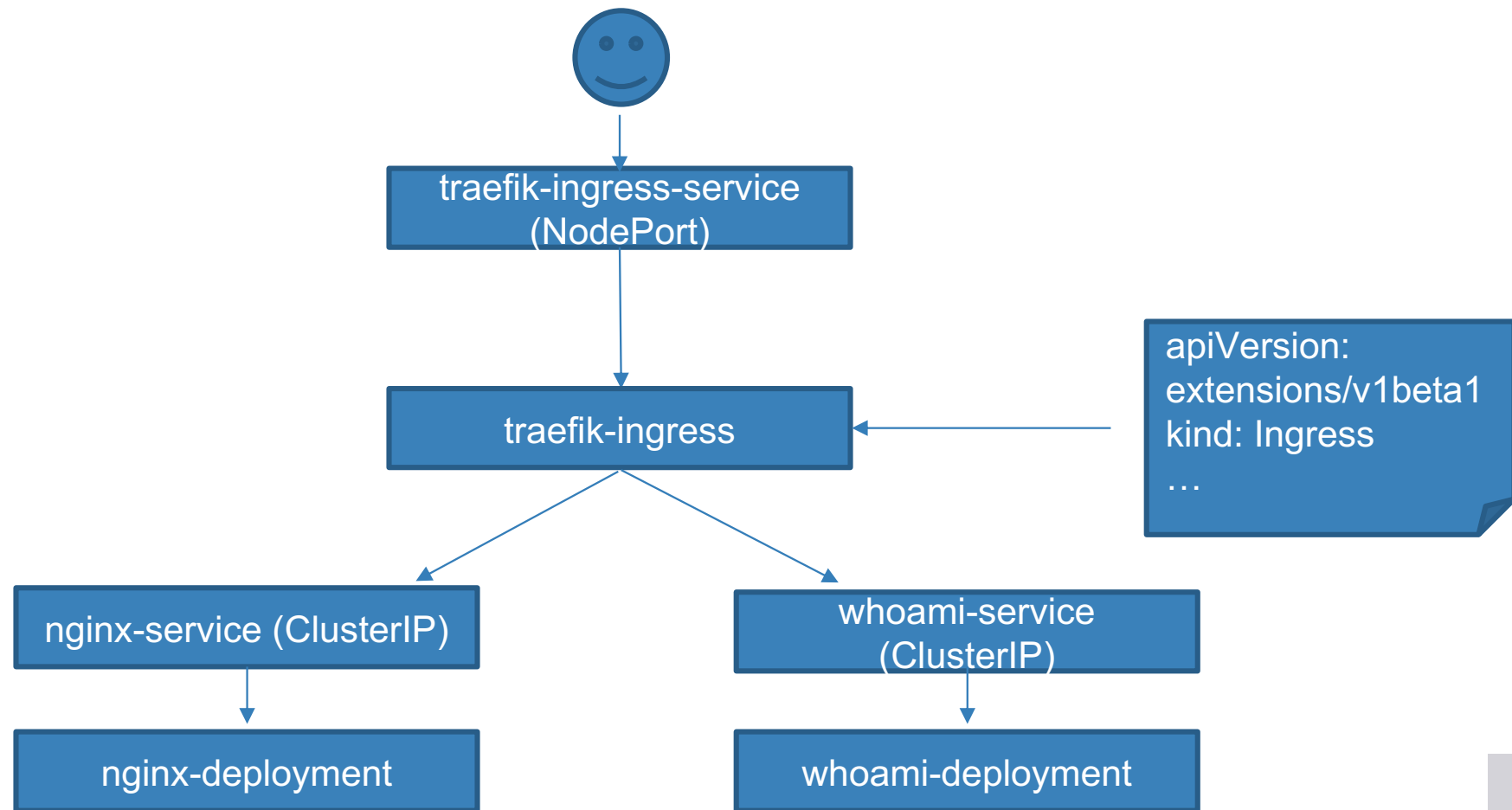
Ingress



Ingress



Lab 24 - Ingress



LAB

https://github.com/sunnyvale-academy/ITS-ICT_Containers

Lab 24 – Ingress

Assignment 07 - Expose Deployments using Ingress